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Machine Learning Algorithms for Intrusion Detection in IoT-enabled Smart Homes

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Abstract

The Internet of Things provides many useful opportunities and makes everyday life easier. At the same time, when interacting with the Internet of Things, a large amount of confidential information is processed, thereby raising the issue of ensuring security. The presented work researches the process of ensuring the security of smart homes by analyzing network traffic to detect malware, which uses machine learning methods Gradient Descent, K-Nearest Neighbor, Recurrent Neural Networks. The research was conducted using the IoT-23 intrusion detection dataset. To evaluate the methods, the metrics accuracy, recall, precision, and F1 were used. As a result, Gradient Descent achieved the highest performance and can be used for intrusion detection.

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1. Introduction

The Internet of Things (IoT) is a network of electronic devices equipped with built-in technologies to interact with each other and the external environment. The concept of IoT has become an integral part of everyday life, providing users with unique opportunities in almost all areas [1]. Today, there are hundreds of millions of connected smart home devices, and Statista predicts that the number of smart homes will exceed 400 million in 2024. Smart homes provide many useful features such as wireless door locks, automatic barometric pressure monitoring, etc. Along with the many advantages of smart homes, the issue of security and privacy arises. Security is associated with the analysis and

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detection of anomalous events in the network traffic of a smart home [2]. Anomaly detection informs about system errors, potential attacks and malicious activities, which can help ensure the smooth and secure operation of the smart home infrastructure. In this regard, one of the most important issues is the study of ways to improve security during the interaction of the IoT in the field of smart home [3, 4].

Machine learning methods are actively used in various fields and at the same time demonstrate promising opportunities for improving the security of smart homes [3, 5]. The interaction of the IoT in the smart home sector generates a large amount of various data, which also complicates the process of reading and processing data. In this regard, an integrated approach is required to ensure the security of a smart home. A large number of studies use different machine learning methods, like SVM, Random Forest, Decision Tree, XGBoost, CNN, etc., which in turn demonstrates high performance in detecting intrusions[5]. The presented research is aimed at detecting anomalous network packets during the interaction of the IoT in the smart home sector using Gradient Descent, K-Nearest Neighbor (KNN), and Recurrent Neural Networks (RNN) machine learning methods. The experimental evaluation was carried out using a real IoT-23 dataset and a systems research methodology was applied, which consists of data pre-processing, feature extraction and training of machine learning models [6].

2. Related Work

There are many works that discuss various methodologies and mechanisms for increasing security and maintaining privacy when interacting with the IoT in the smart home sector. The search for works was carried out in various bibliometric databases, such as IEEE Xplore, Web of Science, ACM DL, Scopus, Google Scholar using the keywords “IoT security”, “Smart Home”, “Malware”, “Machine learning” over the past 5 years. Based on the search results, a content analysis was carried out and 17 works were selected.

Wang M et al. propose a strategy to study the behavior and results of attacks on different types of data that are created in a heterogeneous IoT environment. The proposed method uses a self-service instrument to learn contextual embeddings for input organize highlights [7]. Rani D. et al. proposed an effective intrusion detection system by classifying and predicting various types of network attacks based on network traffic data and real-time sensors of an IoT-enabled smart environment. The system was developed using ensemble algorithms based on machine learning [8]. Visoottiviseth V. et al. explore a home network security system based on the signature method and behavior of IoT devices, as well as machine learning. As a result, the system prevents five main types of attacks [9]. Wang Y. et al. presented a multi-layered security monitoring system that collects, analyzes and characterizes data transmission of heterogeneous IoT devices through programmable home routers. Experimental evaluations have shown that the system is capable of detecting anomalous activity targeting IoT devices in smart homes [10]. A large number of studies demonstrate the growth in the development of smart home infrastructure and, at the same time, show the relevance of the issue of ensuring the security of the IoT.

IoT devices are susceptible to different types of malicious activities that serve different purposes. Table 1 shows commonly encountered malware. The most common types of IoT devices attacked in the smart home are routers and cameras. Routers and cameras contain or have access to personal data.

Table 1. Types of IoT malware.

Malware	Work	Exploit Applied	Smart Home Device	Malware	Work	Exploit Applied	Smart Home Device
Mirai	[11]	Default login credentials	Home Routers, CCTV camera	Hajime	[14]	Default password	Routers, camera
Reaper	[12]	Default password code injection	CCTV cameras, routers	VPNfilter	[15]	Java injection	Routers
Tsunami	[13]	Remote code execution	Routers, camera	Bashlite	[16]	Remote code execution	CCTV cameras, Home Routers

This section provided a brief overview of related work, presenting commonly encountered malware and its detection rate in the smart home industry, thereby highlighting the importance of the ongoing research.

3. Proposed Model

This section first notes the applicability of machine learning methods, then describes the proposed model for intrusion detection in a smart home network and presents descriptions of the Gradient Descent, KNN, RNN methods.

Machine learning methods have great potential in solving security problems. Given the low computing power and memory limitations, the IoT is not able to detect intrusion on every device in the smart home network. It is also important to take into account the life cycle of Internet devices - components of the smart home as a whole.

The presented study proposes a research algorithm that consists of 4 stages that contain machine learning methods. At the initial stage, the input data set is processed, then the data is transferred to the classifier and features are selected. The next stages are training and testing; in the proposed study, machine learning methods such as Gradient Descent, KNN, RNN are used at these stages and as a result of which intrusions will be detected [3, 5]. In general, the proposed study identifies the most suitable machine learning method that will help the IoT as part of a smart home to classify network traffic in real time and thereby detect a malicious flow.

3.1 Gradient Descent

The Gradient Descent algorithm is an optimization algorithm for linear regression problems [3]. The gradient descent algorithm is capable of updating model parameters to achieve the minimum value of the gradient of the loss function, where is the ζ - learning rate. The algorithm function is defined using formula (1).

$$g_{\psi}x = \Psi^M x = \Psi_0 x_0 + \Psi_1 x_1 + \dots + \Psi_n x_n, \quad (1)$$

The loss function for minimizing the cost function is determined using a number of parameters - ψ and is calculated using the formula (2):

$$Q(\Psi) = \frac{1}{2} \sum_{i=1}^k (g_{\psi}(x^{(i)}) - y^{(i)})^2, \quad (2)$$

where $x^{(i)}$ - input feature matrix, $y^{(i)}$ - target matrix, i -number of samples .

The gradient descent algorithm is successfully applied in the study and helps improve feature selection to improve the accuracy of intrusion detection. The table 2 shows the advantages and disadvantages identified during the analysis of different articles. Overall, using the gradient descent method in intrusion detection for the Internet of Things provides benefits such as increased detection accuracy, improved security, and high performance. However, challenges related to complexity, scalability, and energy efficiency remain to be overcome.

3.2 K-Nearest Neighbour

The KNN algorithm could be a administered machine learning strategy utilized to solve classification and regression issues [3]. It could be a non-parametric strategy that produces forecasts based on the similitude of information focused in a given information set. The KNN calculation works by finding the K closest neighbors to a given information point based on a remove metric, Euclidean remove (3):

$$E.D. = \sqrt{\sum (x_i - y_i)^2}. \quad (3)$$

The KNN offers several technical advantages in intrusion detection for IoT, such as simplicity, fast similarity calculation, and high accuracy. However, it also has limitations, including potential problems of overfitting and underfitting, issues related to feature selection, and data quality. Compared to other machine learning algorithms, KNN demonstrates competitive performance in detecting various types of attacks in IoT environments. The table 2 shows the advantages and disadvantages of the KNN algorithm obtained as a result of a literature review

3.3 Recurrent Neural Networks

The RNN are a sort of neural system where associations between components shape a coordinated arrangement [3]. This makes it conceivable to prepare an arrangement of occasions in time or successive spatial chains. RNN uses a feedback loop to model a system where hidden states depend not only on the current observation, but also depend on the previous hidden state (4):

$$\underline{f}_t = \tanh(q_f[k_{t-1}, x_t] + v_f), \quad f_t = q_t * f_{t-1} + i_t * \underline{f}_t, \quad (4)$$

where x_t - input at current timestamp, k_{t-1} - previously state of output, v_f - biases for the respective f , $\underline{f}_t$ - current memory content, q_t - forgot gate, f_{t-1} - previous memory content, f_t - new memory content. The output k_t is presented by the formula (5):

$$k_t = q(k_{t-1}, x_t) = o_t + \tanh(f_t). \quad (5)$$

Using RNN in IoT intrusion detection provides benefits such as capturing temporal relationships, higher detection rates, resilience to dynamic threats, and efficiency at the edge. However, RNNs may have limitations due to computational complexity and information loss which were presented in Table 2. In general, RNNs contribute to the improvement of IoT intrusion detection systems by improving intrusion detection, providing multi-layered attack detection, and improving the security posture of IoT systems.

Table 2. The advantages and disadvantages of ML methods.

Gradient Descent			K-Nearest Neighbour			Recurrent Neural Networks		
Work	Advantages	Disadvantages	Work	Advantages	Disadvantages	Work	Advantages	Disadvantages
[17]	Improved Detection Accuracy	Complexity and Scalability	[20]	High classification efficiency	Sensitive to noise and outliers	[23]	Adaptability to Dynamic Environments	Imbalance in Training Data
[18]	Enhanced Security	Energy Efficiency	[21]	Provides high detection accuracy	Limited scalability	[24]	High Accuracy	Overfitting
[19]	Efficient Model Accuracy	Complexity	[22]	Effective intrusion detection	Computationally Expensive	[25]	Enhanced Security	Lack of Interpretability

4. Experimental results

This section describes the hardware and software environment for the experimental study, and also presents the stages of preparing the dataset and metrics for evaluating the algorithms.

The research was conducted on a personal computer HP, with the operating system Windows 10 Enterprise (64 bits), processor Intel(R) Core(TM). Processor i5-10500@ 3.6 GHz, RAM- 16.0 GB, video card - NVIDIA GeForce GTX 1070. Experimental studies were conducted using the IOT-23 network traffic dataset, which has 20 malware interceptions performed on IoT devices and 3 interceptions of benign IoT device traffic [4].

In the IOT-23 dataset, null values were removed and one variable was isolated that contains the network traffic label. The data set was divided into training (80%) and test (20%) parts. Then the models were trained using Gradient Descent, KNN, RNN methods. To evaluate the models under study, the metrics shown in Table 3 were used, taking into account that F(true-positive), B(true-negative), C(false-positive), D(false-negative).

Figure 1 shows the results of the machine learning methods that were used in this study. A comparison of the proposed methods was carried out to detect intrusions in a smart home network, as a result of which Gradient Descent demonstrated a better performance compared to KNN and RNN. Gradient Descent, KNN and RNN achieved 73,23%, 65,53% and 66,94% accuracy.

Table 3. Metrics for evaluating models.

Nomenclature	Metric	Equation	Nomenclature	Metric	Equation
A	accuracy	$(F + B)/(F + B + C + D)$	P	precision	$F/(F + C)$
R	recall	$F/(F + \Omega)$	F1	f1-score	$2F/(2F + C + D)$

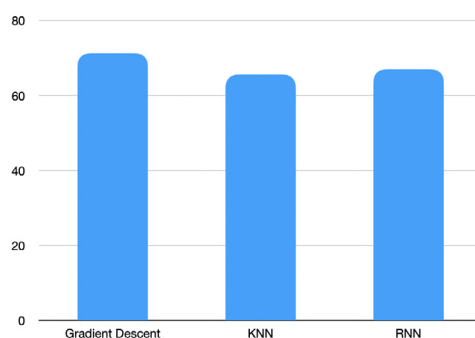


Fig. 1. Results of machine learning methods in intrusion detection.

The table 4 demonstrates research works by different authors in the field of machine learning, detailing the datasets used, algorithms applied, and the corresponding evaluation metrics. The presented studies were selected to compare the presented model with other similar approaches.

Table 4. Comparisons with other work.

Work	Dataset	Algorithm	Metrics	Work	Dataset	Algorithm	Metrics
[26]	IoT-23	Decision Tree	Accuracy 0,73	[27]	IoT-23	KNN	F1 0,60
[28]	BoT-IoT	Gradient Boosting	Accuracy 0,99	Proposed Model	IoT-23	Gradient Descent	Accuracy 0,73

5. Conclusion

Along with the growth of IoT, the issue of security is also gaining momentum. The process of detecting attacks and malware is becoming one of the complex tasks associated with the large volume of diverse data generated by IoT devices. To solve this problem, machine learning methods are actively used, which involve a learning and decision-making process, which allows making predictions. Three machine learning methods were considered in the study: Gradient Descent, KNN and RNN. Based on the experimental data, it should be noted that the Gradient Descent algorithm is superior to the other two considered algorithms for the purpose of detecting and classifying anomalies in the context of the IoT-23 data set, which highlights its applicability. The relatively low results of the studied algorithms demonstrate the possibility of using them in detecting low-frequency intrusions. It is important to note that the study was conducted on a personal computer with average characteristics, and therefore the results obtained are of an exploratory nature. In the future, we plan to consider this problem using more powerful computing resources and using other popular machine learning methods.

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